8.4 Data Displays

Name _____ Period ____ Date _____

Algebra I 8.4 (30-minute plan) · Student sheet

I can…

- tell whether data are qualitative or quantitative.
- choose an appropriate display for a data set and explain why.
- read and interpret common data displays.

Vocabulary · fill in the missing word

Qualitative (categorical) data — data that are labels or _____, such as colors, names, or zip codes

Quantitative data — numbers that are _____ or counted, such as heights or test scores

Bar graph — compares the counts of _____; the bars don't touch

Circle graph — shows parts of a _____; each section's angle is its share of 360°

Histogram — shows how many _____ values fall in each interval; the bars touch

Line graph — shows how data _____ over time

Box-and-whisker plot — shows the _____ summary and spread of numerical data

Part 1 · Notes and examples (fill in each note, then work the example)

Qualitative data are _____; **quantitative** data are numbers that are _____ or _____.

Example 1

Tell whether the data are qualitative or quantitative.

a. favorite colors of students

b. heights of plants

c. zip codes of customers

d. numbers of pets owned

Choosing a display: Categories with counts → _____ Parts of a whole → _____

Choosing a display: Numbers in intervals → _____ Change over time → _____

Example 2

Which display best represents each data set?

a. a city's monthly rainfall over one year

b. the percent of a budget spent on each type of expense

c. the number of students choosing each lunch option

d. the median and spread of the prices of 40 laptops

Circle graphs: In a circle graph, a section's angle = its share · _____.

Example 3

The table shows the favorite seasons of 30 students.

a. What percent of the students chose fall?

b. In a circle graph, what is the central angle of the fall section?

c. Name two good displays for these data.

Season	Spring	Summer	Fall	Winter
Students	9	12	6	3

Part 2 · Partner practice

Partner practice

Tell whether the data are qualitative or quantitative: the number of text messages sent in a day, favorite type of music, area codes, weights of backpacks.